

Implementation

Database Setup

Create / Use Database

```
mysql> CREATE DATABASE OrderDB;  
mysql> USE OrderDB;
```

Database changed

Table Creation

Create Customers Table

```
mysql> CREATE TABLE Customers (  
->     CustomerID INT PRIMARY KEY,  
->     Name        VARCHAR(50),  
->     Email        VARCHAR(50),  
->     City         VARCHAR(50)  
-> );
```

Query OK, 0 rows affected (0.02 sec)

Create Orders Table

```
mysql> CREATE TABLE Orders (  
->     OrderID      INT PRIMARY KEY,  
->     CustomerID   INT,  
->     OrderDate    DATE,  
->     Amount       INT  
-> );
```

Query OK, 0 rows affected (0.02 sec)

Inserting Data

Insert into Customers Table

```
mysql> INSERT INTO Customers VALUES  
-> (1, 'Ramesh', 'ramesh@gmail.com', 'Pune'),  
-> (2, 'Amit',   'amit@gmail.com',   'Mumbai'),  
-> (3, 'Sneha',  'sneha@gmail.com',  'Nashik'),  
-> (4, 'Rahul',  'rahul@gmail.com',  'Nagpur'),  
-> (5, 'Priya',  'priya@gmail.com',  'Delhi');
```

Query OK, 5 rows affected (0.00 sec)

Insert into Orders Table

```
mysql> INSERT INTO Orders VALUES  
-> (101, 1, '2024-01-10', 5000),  
-> (102, 2, '2024-01-12', 7000),
```

```
-> (103, 1, '2024-01-15', 3000),  
-> (104, 6, '2024-01-18', 4000); -- CustomerID 6 not in Customers
```

```
Query OK, 4 rows affected (0.00 sec)
```

Display Table Data

View Customers Table

```
mysql> SELECT * FROM Customers;
```

```
+-----+-----+-----+-----+  
| CustomerID | Name   | Email                | City   |  
+-----+-----+-----+-----+  
|          1 | Ramesh | ramesh@gmail.com    | Pune   |  
|          2 | Amit   | amit@gmail.com       | Mumbai |  
|          3 | Sneha  | sneha@gmail.com      | Nashik |  
|          4 | Rahul  | rahul@gmail.com      | Nagpur |  
|          5 | Priya  | priya@gmail.com      | Delhi  |  
+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

View Orders Table

```
mysql> SELECT * FROM Orders;
```

```
+-----+-----+-----+-----+  
| OrderID | CustomerID | OrderDate | Amount |  
+-----+-----+-----+-----+  
|      101 |          1 | 2024-01-10 | 5000 |  
|      102 |          2 | 2024-01-12 | 7000 |  
|      103 |          1 | 2024-01-15 | 3000 |  
|      104 |          6 | 2024-01-18 | 4000 |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

SQL Join Queries

Query 1: INNER JOIN

Retrieve all customer orders (only customers who have placed orders)

```
mysql> SELECT c.CustomerID, c.Name, o.OrderID, o.Amount  
-> FROM Customers c  
-> INNER JOIN Orders o ON c.CustomerID = o.CustomerID;
```

```
+-----+-----+-----+-----+  
| CustomerID | Name   | OrderID | Amount |  
+-----+-----+-----+-----+  
|          1 | Ramesh |      101 | 5000 |  
|          2 | Amit   |      102 | 7000 |  
|          1 | Ramesh |      103 | 3000 |
```

```
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Query 2: LEFT JOIN

Retrieve all customers, including those who have not placed any orders

```
mysql> SELECT c.CustomerID, c.Name, o.OrderID, o.Amount
-> FROM Customers c
-> LEFT JOIN Orders o ON c.CustomerID = o.CustomerID;
```

```
+-----+-----+-----+-----+
| CustomerID | Name   | OrderID | Amount |
+-----+-----+-----+-----+
|          1 | Ramesh |      103 |    3000 |
|          1 | Ramesh |      101 |    5000 |
|          2 | Amit   |      102 |    7000 |
|          3 | Sneha  |      NULL |     NULL |
|          4 | Rahul  |      NULL |     NULL |
|          5 | Priya  |      NULL |     NULL |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Query 3: RIGHT JOIN

Retrieve all orders, including those placed by customers not in the Customers table

```
mysql> SELECT c.CustomerID, c.Name, o.OrderID, o.Amount
-> FROM Customers c
-> RIGHT JOIN Orders o ON c.CustomerID = o.CustomerID;
```

```
+-----+-----+-----+-----+
| CustomerID | Name   | OrderID | Amount |
+-----+-----+-----+-----+
|          1 | Ramesh |      101 |    5000 |
|          2 | Amit   |      102 |    7000 |
|          1 | Ramesh |      103 |    3000 |
|        NULL | NULL   |      104 |    4000 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Query 4: FULL OUTER JOIN

Retrieve a complete list of customers and orders, including those with no match in the other table

```
mysql> SELECT c.CustomerID, c.Name, o.OrderID, o.Amount
-> FROM Customers c
-> LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
-> UNION
-> SELECT c.CustomerID, c.Name, o.OrderID, o.Amount
-> FROM Customers c
-> RIGHT JOIN Orders o ON c.CustomerID = o.CustomerID;
```

```
+-----+-----+-----+-----+
| CustomerID | Name   | OrderID | Amount |
+-----+-----+-----+-----+
|          1 | Ramesh |      103 |   3000 |
|          1 | Ramesh |      101 |   5000 |
|          2 | Amit   |      102 |   7000 |
|          3 | Sneha  |      NULL |   NULL |
|          4 | Rahul  |      NULL |   NULL |
|          5 | Priya  |      NULL |   NULL |
|        NULL | NULL   |      104 |   4000 |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```